

Non-Linear Dynamics Notes

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Lecture 1

1.1 Books

- Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed., Westview Press, 2015.

This is Bible for the subject. It is standard, detailed and a bit old text.

1.2 Dynamics

- Study of evolution of systems
Eg. Simple Pendulum has equation of motion

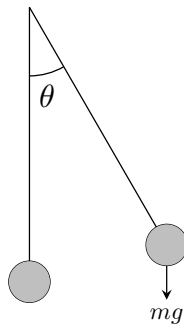


Figure 1.1:

$$\ddot{\theta} - \frac{g}{l} \sin \theta = 0 \quad (1.1)$$

For small angles

$$\ddot{\theta} - \frac{g}{l} \theta = 0 \quad (1.2)$$

Which is linear. But if θ is not small then non-linear and hence included in NLD¹.

- As equation becomes Non-Linear, the solution drastically changes
- Non-Negligible Non-Linear terms are not perturbative and needs separate study
- The subject NLD has advanced after the computers became powerful enough to perform high level computation

1.3 1D Maps

- Suppose we have a sequence of numbers (real numbers mostly)

$$x_0, x_1, x_2, \dots$$

Such that

$$x_0, x_1 = f(x_0), x_2 = f(x_1), \dots \quad (1.3)$$

i.e.

$$x_{n+1} = f(x_n) \quad (1.4)$$

Then it is called an *Map*²³

1.3.1 Fixed Point

- Solution of (1.3) such that

$$x^* = f(x^*) \quad (1.5)$$

then x^* ⁴ is called *fixed point*. Note that it will give rise to the sequence of form

$$x_0, x_1, x_2, \dots, x^*, x^*, x^*, \dots$$

1.3.2 Stability of a Fixed Point

- Suppose $x_0 = \bar{x} + \delta$.

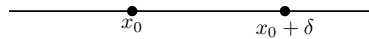


Figure 1.2:

¹**NLD:** Non-Linear Dynamics

²Note here that the function f remains same

³If the function f has random number in the function itself then it is *Undeterministic Map*. We'll consider only *Deterministic Maps*

⁴It is widely followed conventions to label fixed point as x^* and we'll continue using it

- Now if the subsequent x 's approach the fixed point x^* then the fixed point is stable and called an *attractor*

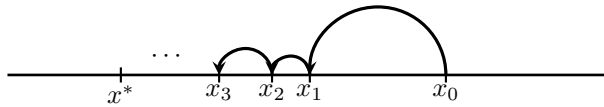


Figure 1.3: Attractor

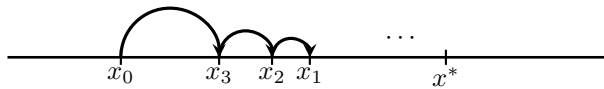


Figure 1.4: Attractor

- Approach to x^* is Oscillatory and is still fixed point

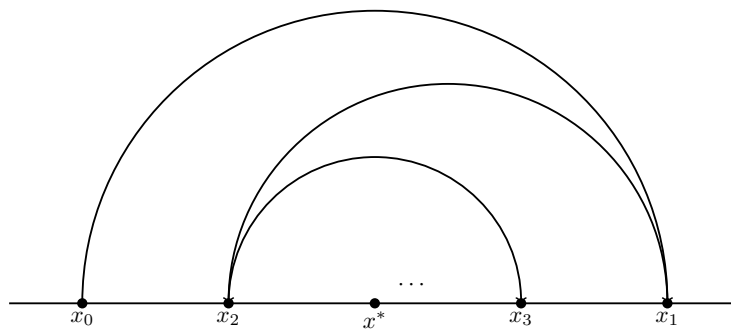


Figure 1.5: Oscillatory Approach

- If the subsequent points diverge from the fixed point (monotonically or oscillatory) then the point x^* is Unstable Fixed Point

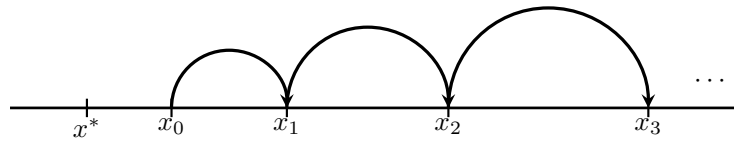


Figure 1.6: Repeller

1.3.3 Digramatic way to find number of fixed points

Follow the steps:

1. Draw the line $y = x$
2. Draw the curve $y = f(x)$
3. No. of intersections = No. of Fixed Points⁵

Example 1.1. $f(x) = \sin x$

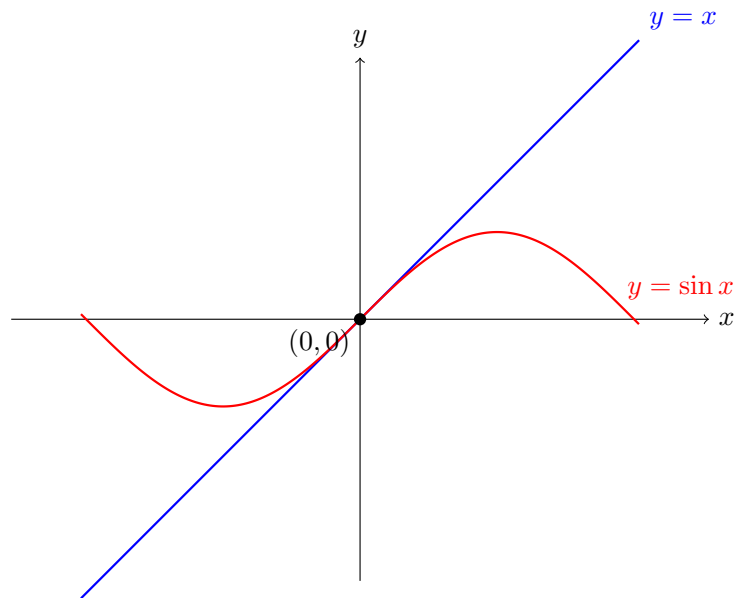


Figure 1.7:

There is only one fixed point at $x = 0$

Example 1.2. $f(x) = \cos x$

There is only one fixed point at $x \approx 0.74$

⁵This logic is clear from the definition of the fixed point as (1.3.1)

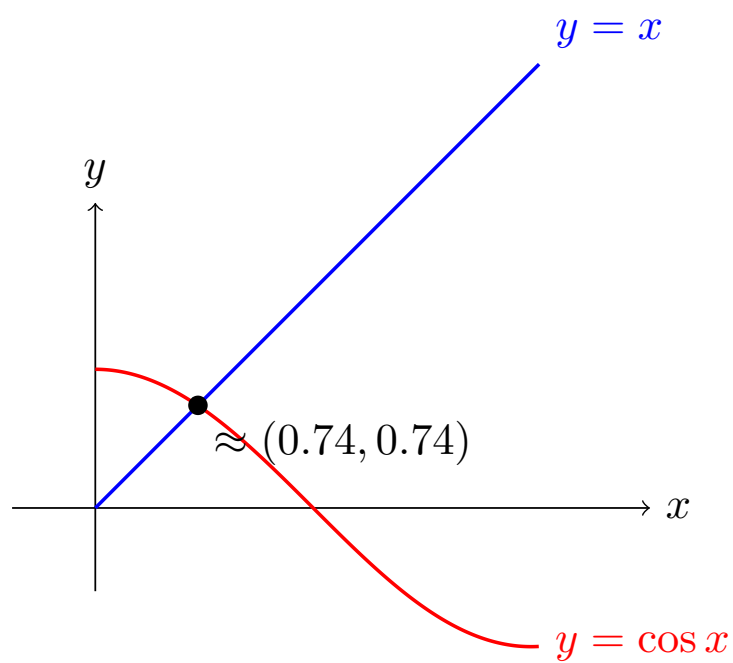


Figure 1.8:

Example 1.3. $f(x) = \frac{2x}{1+x}$

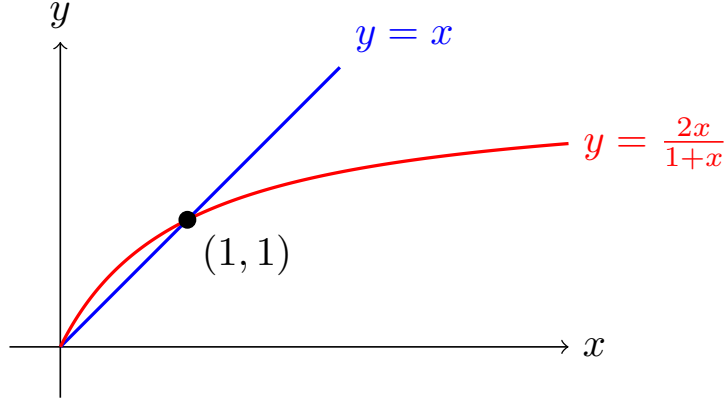


Figure 1.9:

There is only one fixed point at $x = 1$

- In monotonic approach

$$|x_{n+1} - x^*| < |x_n - x^*| \implies \text{Stable fixed point}$$

$$|x_{n+1} - x^*| > |x_n - x^*| \implies \text{Unstable fixed point}$$

- We define the quantity *multiplier* by

$$\mu = \frac{x_1 - x^*}{x_0 - x^*} \quad (1.6)$$

It determines whether the fixed point is stable or unstable.

$$|\mu| < 1 \implies \text{Stable} \quad (1.7)$$

$$|\mu| > 1 \implies \text{Unstable} \quad (1.8)$$

$$\mu > 0 \implies \text{Monotonic} \quad (1.9)$$

$$\mu < 0 \implies \text{Oscillatory} \quad (1.10)$$

- Now, let's analyze the fixed point analytically. Let

$$x_0 = x^* + \delta \quad (1.11)$$

Then

$$\begin{aligned} x_1 &= f(x_0) = f(x^* + \delta) \\ &= f(x^*) + \delta f'(x^*) + \frac{1}{2} \delta^2 f''(x^*) + \dots \\ &= x^* + \delta f'(x^*) + \frac{1}{2} \delta^2 f''(x^*) + \dots \end{aligned}$$

- Now linearize

$$x_1 = x^* + \delta f'(x^*) \quad (1.12)$$

Therefore,

$$f'(x^*) = \frac{x_1 - x^*}{\delta} = \frac{x_1 - x^*}{x_0 - x^*}$$

Using (1.6)

$$\mu = f'(x^*) \quad (1.13)$$

- Now in Example 1.1: $\mu = \cos x^* = 1$. The case $\mu = 1$ ⁶ needs special analysis and we skip it for a moment
- In Example 1.2: $\mu = -\sin(x^*) < 0$, hence fixed point is stable and oscillatory.
- In Example 1.3:

$$\mu = \frac{2(1+x^*) - 2x^*}{(1+x^*)^2} = \frac{2}{(1+x^*)^2}$$

At $x^* = 0$; $\mu = 2 \implies$ monotonically unstable

At $x^* = 1$; $\mu = 1/2 \implies$ monotonically stable

- Calculating Stability form multiplier is only valid when Linearization is valid.
- We now adopt Diagramatic Approach: *Cobwebb Diagrams*
 1. Draw curve $y = x$
 2. Draw $y = f(x)$
 3. Draw line from x_0 to $f(x)$
 4. Draw horizontal line to $y = x$ and that point of intersection gives x_1

⁶In this case the fixed point x^* is called *Marginally Stable*

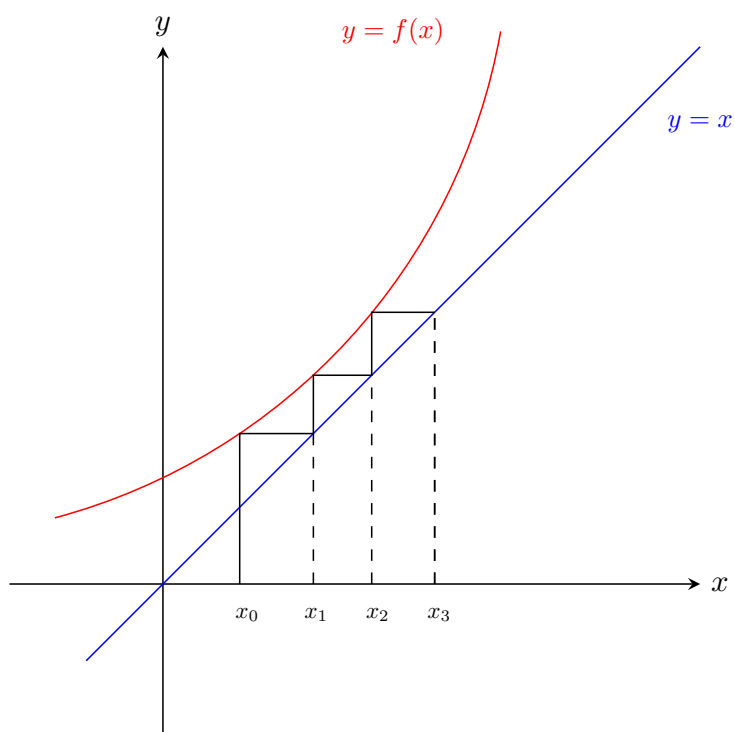


Figure 1.10:

The figure above depicts an example.

Now let's draw Cobweb diagrams to our three examples.

- For Example [1.1](#)

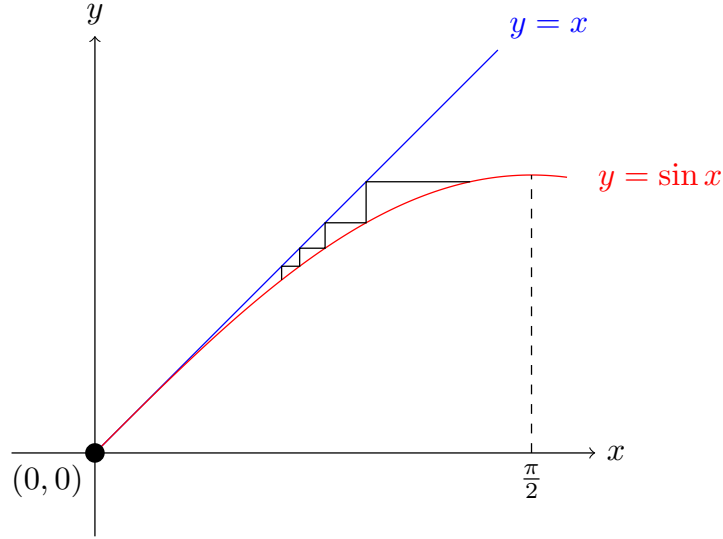


Figure 1.11:

Note that Cobweb diagram shows $x = 0$ as stable fixed point unlike our analytical approach which failed due to $\mu = 1$.

- For Example 1.2

$\mu = 0$ and $\mu = 1$ Case:

$$x_1 = x^* + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \dots$$

- for $\mu = 1$

$$\begin{aligned} x_1 &= x^* + \delta + \frac{1}{2}\delta^2 f''(x^*) + \dots \\ &= x_0 + \frac{1}{2}\delta^2 f''(x^*) + \dots \end{aligned} \tag{1.14}$$

hence from above the higher order terms are important and will determine stability.

- For $\mu = 0$

$$\begin{aligned} x_1 &= x^* + \frac{1}{2}\delta^2 f''(x^*) + \dots \\ &\approx x^* + c\delta^2 \quad \left\{ c = \frac{1}{2}f''(x^*) \right\} \end{aligned} \tag{1.15}$$

Now,

$$x_2 = x^* + c(c\delta^2)$$

And hence,

$$x_n = x^* + c(\dots)\delta^{2^n} \tag{1.16}$$

Since convergence is very fast, it is *superstable* fixed point.

•

$$\begin{aligned} x_1 &= f(x_0) = f(x^* + \delta) \\ &= x^* + \delta f'(x^*) + \frac{1}{2} \delta^2 f''(x^*) + \dots \end{aligned}$$

Linearize

$$\begin{aligned} x_1 &= x^* + \delta\mu \\ x_2 &= f(x_1) = f(x^* + \delta\mu) \\ &= x^* + \mu^2\delta \\ x_3 &= \mu^3\delta \\ x_n &= x^* + \mu^3\delta \end{aligned} \tag{1.17}$$

$$\begin{aligned} \frac{x_n - x^*}{x_0 - x^*} &= \mu^n = e^{n \log \mu} \quad (\mu > 0) \\ &= (|\mu| \text{sign}(\mu))^n \\ &= [\text{sign}(\mu)]^n e^{n \log |\mu|} \\ x_n &= x^* + \delta \cdot (\text{sign}(\mu))^n e^{n \log |\mu|} \end{aligned} \tag{1.18}$$

1.4 Logistic Map

We define *Logistic Map*⁷⁸ by

$$f(x) = rx(1 - x) \quad r > 0 \tag{1.19}$$

- Now let's find the fixed points

$$\begin{aligned} x^* &= rx^*(1 - x^*) \\ \boxed{x^* = 0} \end{aligned} \tag{1.20}$$

$$\begin{aligned} 1 &= r(1 - x^*) \\ \boxed{x^* = 1 - \frac{1}{r}} \end{aligned} \tag{1.21}$$

⁷ $f(x)$ is quadratic hence included in NLD

⁸This equations appears in many areas: Engineernig, Population Dynamics, etc.

- We'll always keep $0 \leq |x_n| \leq 1$ and vary $r > 0$
 $x^* = 0$ exist for all r
 $x^* = 1 - \frac{1}{r}$ exist for $r \geq 0$
- Stability of Fixed Points

$$f'(x) = r - 2rx$$

At $x^* = 0$ $f(x^*) = r \implies$ stable fixed point for $r < 1$ and unstable for $r > 1$; $r = 1$ is marginally stable
 At $x^* = 1 - \frac{1}{r}$

$$\begin{aligned} f(x^*) &= r - 2r \left(1 - \frac{1}{r}\right) \\ &= r - 2r + 1 \\ &= 2 - r \end{aligned}$$

for $r = 1$ $\mu = -1$ marginally stable
 for $r = 3$ $\mu = 1$ marginally stable

- for $r > 3$ (say $r = 3.1$) we get sequence

$$x_0, x_1, \dots, p, q, p, q, \dots \quad (1.22)$$

This alternate sequence of p 's and q 's is called a *2-cycle*⁹.

⁹A fixed point is a 1-cycle

Lecture 2

2.1 Logistic Map (cont...)

- For stability

$$\begin{aligned} -1 < \mu < 1 &\implies -1 < 2 - r < 1 \\ &\implies -3 < -r < -1 \\ &\implies 1 < r < 3 \end{aligned}$$

- for a 2-cycle

$$\begin{aligned} f(p) &= q & f(q) &= p \\ \therefore f(f(x)) &= f(x) = x \end{aligned} \tag{2.1}$$

and solution must be p and q

$$\begin{aligned} r(rx(1-x))(1-rx(1-x)) &= x \\ r^2x(1-x)(1-rx(1-x)) &= x \end{aligned}$$

$f(x^*) = x^*$ and therefore $f(f(x^*)) = f(x^*) = x^*$

Therefore, $0, 1 - \frac{1}{r}$ are solution of above equation

$$x \left[r^2(1-x)(1-rx(1-x)-1) \right] = 0$$

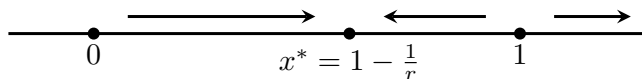


Figure 2.1:

taking out $x = 0$ factor

$$r^2(1-x)(1-rx(1-x)-1)=0$$

$$\text{put } r(1-x)=y \implies rx=r-y$$

$$ry(1-xy)-1=0$$

$$yr-y^2rx-1=0$$

$$yr-y^2(r-y)-1=0$$

$$(y-1)(-yr+y^2+y+1)=0$$

$$y=1 \implies r(1-x)=1 \implies x=1-\frac{1}{r}$$

$$y^2+y(1-r)+1=0 \tag{2.2}$$

Now, consider

$$\begin{aligned} y &= \frac{-(1-r) \pm \sqrt{(1-r)^2 - 4}}{2} \\ &= \frac{(r-1) \pm \sqrt{r^2 - 2r - 3}}{2} \\ &= \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2} \\ r(1-x) &= \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2} \\ x &= 1 - \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2r} \end{aligned}$$

$$x = \frac{1}{2r} \left(r+1 \pm \sqrt{(r-3)(r+1)} \right) \tag{2.3}$$

•

$$p+q = \frac{r+1}{r} = 1 + \frac{1}{r} \tag{2.4}$$

$$\begin{aligned} pq &= \frac{1}{4r^2} \left[(r+1)^2 - (r+1)(r-3) \right] \\ &= \frac{1}{4r^2} (r^2 + 2r + 1 - r^2 + 2r + 3) \\ &= \frac{1}{4r^2} (4r + 4) \end{aligned}$$

$$pq = \frac{1}{r} \left(1 + \frac{1}{r} \right) \tag{2.5}$$

- The 2-cycle exists only for $r > 3$
 $x^* = 1 - \frac{1}{r}$ stable for $1 < r < 3$
- Suppose we start with the point $p + \delta_0$, then we'll get sequence

$$p + \delta_0, p + \delta_1, p + \delta_2, \dots$$

Question: Does $\delta_n \rightarrow 0$

$$\begin{aligned} \mu_2 = \left| \frac{x_2 - p}{x_0 - p} \right| < 1 &\implies \text{We approach gradually to } p; \text{ stable 2-cycle} \\ &> 1 \implies \text{unstable 2-cycle} \end{aligned}$$

•

$$\begin{aligned} x_2 &= f^2(x_0) \\ &= f^2(p + \delta) \\ &= f^2(p) + \delta \frac{d^2}{dx^2} f^2(p) \Big|_{x=p} \end{aligned}$$

Write $g = f(x) \implies g(p) = f(p) = p$ and $f^2(x) = f(g)$

$$\begin{aligned} \mu_2 &= \frac{d^2}{df(x)^2} f^2(x) \frac{df(x)}{dx} \Big|_{x=p} \\ \mu_2 &= \frac{df(g)}{dg} \frac{df(x)}{dx} \Big|_{x=p} \end{aligned} \tag{2.6}$$

$$\begin{aligned} \mu_2 &= (r - 2rg)(r - 2rx) \\ &= r^2(1 - 2g)(1 - 2x) \Big|_{x=p} \end{aligned} \tag{2.7}$$

let $g = f(x) \quad q = f(p)$

$$\begin{aligned} \mu_2 &= r^2(1 - 2q)(1 - 2p) \\ &= r^2 \left[1 - 2 \left(1 + \frac{1}{r} \right) + \frac{4}{r} \left(1 + \frac{1}{r} \right) \right] \\ &= r^2 \left(-1 + \frac{2}{r} + \frac{4}{r^2} \right) \\ &= 4 + 2r + r^2 \\ &= 5 - (r^2 - 2r + 1) \end{aligned}$$

$$\mu_2 = 5 - (r - 1)^2 \tag{2.8}$$

$$\begin{aligned} |\mu_2| < 1 &\quad (\text{Condition for stability of 2-cycle}) \\ \implies -1 < \mu_2 < 1 \\ \implies -1 < 5 - (r - 1)^2 < 1 \\ \implies -6 < -(r - 1)^2 < -4 \\ \implies 4 < (r - 1)^2 < 6 \end{aligned}$$

$$\boxed{3 < r < 1 + \sqrt{6}} \quad (2.9)$$

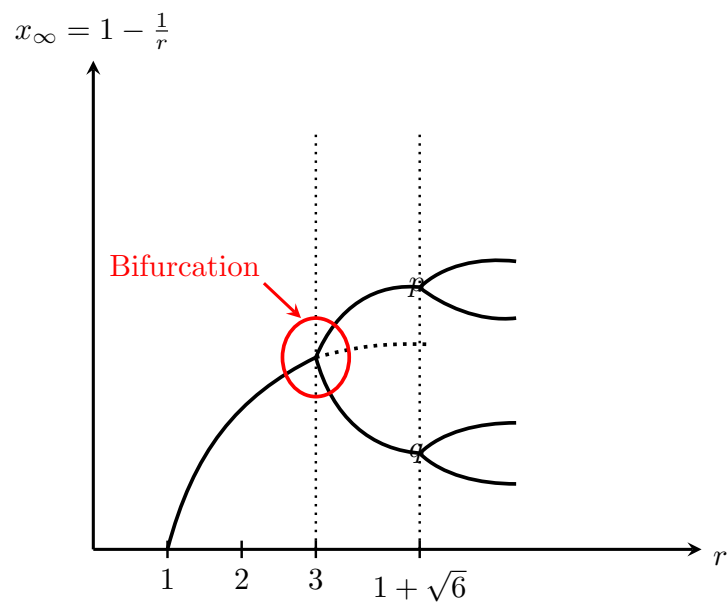
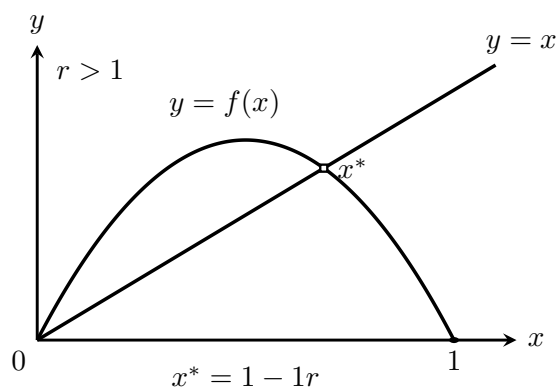


Figure 2.2: Feigenbaum Diagram



Slope to the curve at x^* determines if point is stable or not.

If the slope > 1 unstable and if slope < 1 then stable

Figure 2.3:

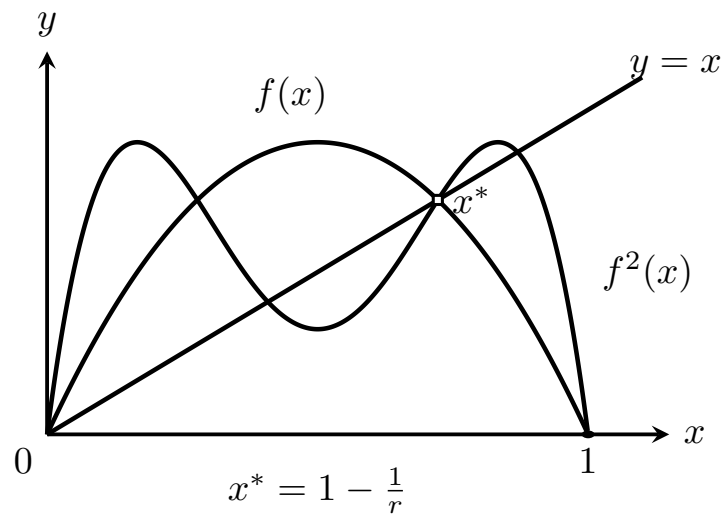


Figure 2.4:

- **Bifurcation:** A fixed point splits in two¹
- $1 - \frac{1}{r}$ exists for $r \leq 1$ but stable for $1 < r < 3$
2-cycle p, q exists for $r > 3$ but stable for $3 < r < 1 + \sqrt{6}$
- for $r > 1 + \sqrt{6}$ we get

$$x_0, x_1, x_2, \dots, p, q, r, s, p, q, r, s, \dots \quad (2.10)$$

which is a 4-cycle²

¹Refer to *H. G. Schuster and W. Just, *Deterministic Chaos*, 3rd ed., Wiley-VCH* for nice diagrams

²As my professor said promotion from bicycle to motorcycle

Lecture 3

3.1 Properties of Logistic Map

- We define the quantity μ_2

$$\mu_2 = \left. \frac{d^2}{dx^2} f^2(x) \right|_{p,q} \quad (3.1)$$

p and $q \implies$ real for $r > 3$, do not exist for $r < 3$
 $|\mu_2| < 1$ for $r < 1 + \sqrt{6} = 3.45$

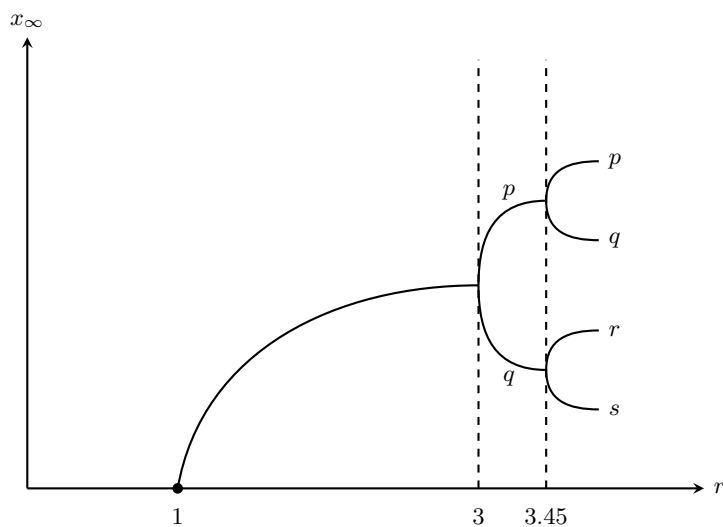


Figure 3.1:

Lecture 6

4.1 2D Maps

- start with pair of numbers

$$\begin{pmatrix} x_n \\ y_n \end{pmatrix} \\ \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}, \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \dots, \begin{pmatrix} x_n \\ y_n \end{pmatrix}, \dots \quad (4.1)$$

- where the mapping functions are

$$x_{n+1} = f(x_n) \quad (4.2)$$

$$y_{n+1} = g(y_n) \quad (4.3)$$

- Fixed points are solution of equations

$$x^* = f(x^*, y^*) \quad y^* = g(x^*, y^*) \quad (4.4)$$

so that

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}, \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \dots, \begin{pmatrix} x^* \\ y^* \end{pmatrix}, \begin{pmatrix} x^* \\ y^* \end{pmatrix}, \begin{pmatrix} x^* \\ y^* \end{pmatrix}, \dots \quad (4.5)$$

- for stability of fixed points consider

$$x_0 = x^* + \Delta x \quad y_0 = y^* + \Delta y$$

-

$$\begin{aligned} x_1 &= f(x^* + \Delta x, y^* + \Delta y) \\ &= f(x^*, y^*) + \Delta y \frac{\partial f}{\partial y} \Big|_{(x^*, y^*)} + \Delta x \frac{\partial f}{\partial x} \Big|_{(x^*, y^*)} \\ &= x^* + f_y \Delta y + f_x \Delta x \end{aligned}$$

similarly

$$y_1 = y^* + g_y \Delta y + g_x \Delta x$$

- we can write this in matrix form

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} x^* + \Delta x \\ y^* + \Delta y \end{pmatrix}$$

then

$$\begin{aligned} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} &= \begin{pmatrix} x^* + f_x \Delta x + f_y \Delta y \\ y^* + g_x \Delta x + g_y \Delta y \end{pmatrix} \\ \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} &= \begin{pmatrix} x^* \\ y^* \end{pmatrix} + \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}_{(x^*, y^*)} \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \\ &= \begin{pmatrix} x^* \\ y^* \end{pmatrix} + J \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \end{aligned} \tag{4.6}$$

- To get next successive points

$$\begin{aligned} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} &= \begin{pmatrix} x^* \\ y^* \end{pmatrix} + J^2 \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \\ &\vdots \\ \begin{pmatrix} x_n \\ y_n \end{pmatrix} &= \begin{pmatrix} x^* \\ y^* \end{pmatrix} + J^n \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \end{aligned}$$

- Suppose J has eigenvalues λ_1, λ_2 with eigenvectors $|e_1\rangle, |e_2\rangle$ respectively, then,

$$\begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} = \Delta\alpha |e_1\rangle + \Delta\beta |e_2\rangle \tag{4.7}$$

$$\begin{pmatrix} x_n \\ y_n \end{pmatrix} - \begin{pmatrix} x^* \\ y^* \end{pmatrix} = J^n \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} = \Delta\alpha \lambda_1^n |e_1\rangle + \Delta\beta \lambda_2^n |e_2\rangle \tag{4.8}$$

- **Case 1:** λ_1, λ_2 are real and $\in (-1, 1) \implies$ stable fixed point
- **Case 2:** one or both λ_1, λ_2 are $> 1 \implies$ unstable fixed point
- **Case 3:** $\lambda_1, \lambda_2 = re^{\pm i\theta}$
 if $r > 1 \implies$ unstable fixed point
 if $r < 1 \implies$ stable fixed point
 if $r = 1 \implies \lambda_1^n, \lambda_2^n = e^{\pm in\theta}$
 $\begin{pmatrix} x_n \\ y_n \end{pmatrix}$ will never grow from $\begin{pmatrix} x^* \\ y^* \end{pmatrix}$

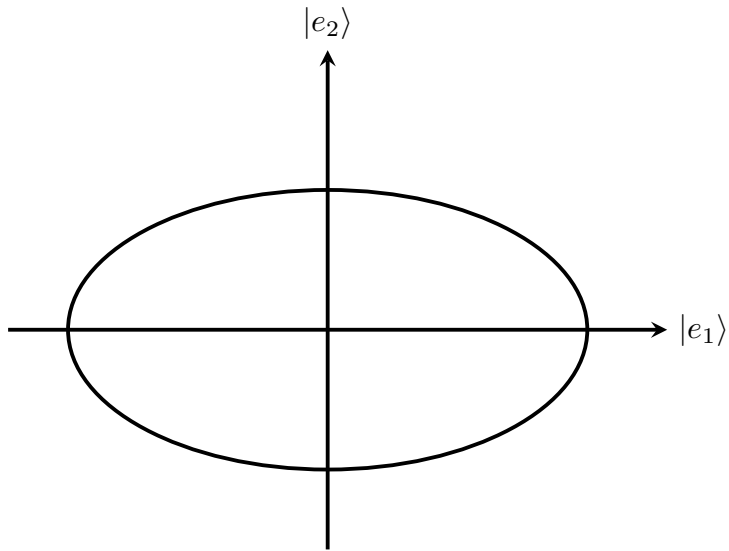


Figure 4.1: Elliptic Fixed Points

- λ_1, λ_2 are real and $\lambda_1 > 1, \lambda_2 < 1$

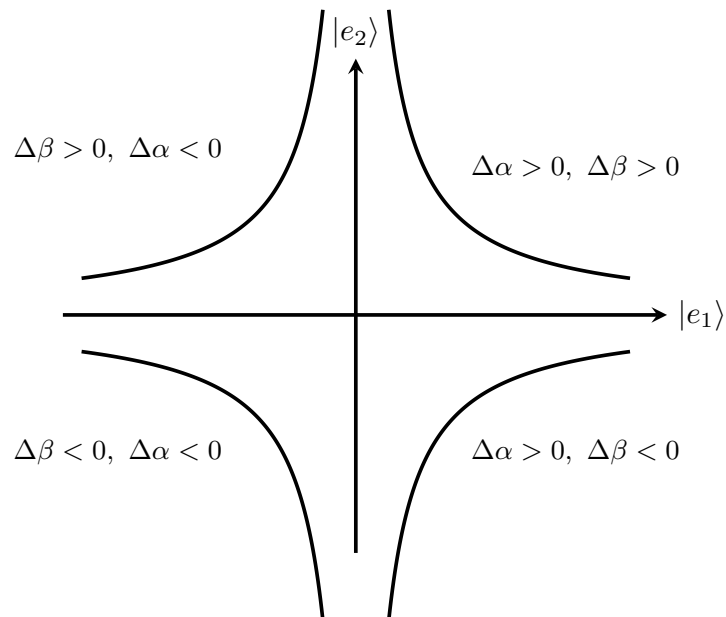


Figure 4.2: Hyperbolic Fixed Point

4.2 Heron Map

- Heron Map is defined by

$$\begin{aligned}x_{n+1} &= 1 - ax_n^2 - y_n \\y_{n+1} &= x_n \\ax^2 + 2x - 1 &= 0\end{aligned}$$

- Fixed points

$$\begin{aligned}x &= 1 - ax^2 - y \\y &= x\end{aligned}$$

giving two fixed points

$$x, y = \frac{-1 \pm \sqrt{1+a}}{a} \tag{4.9}$$

- let $x^* = 1 + \sqrt{1+a}$

$$x^* = \frac{1}{2a} [-c, c - 2] \tag{4.10}$$

Lecture 7

5.1 2 cycles in Heron Map

- The 2-cycle in a 2D map will follow the following sequence

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}, \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \dots, \begin{pmatrix} A_0 \\ B_0 \end{pmatrix}, \begin{pmatrix} A_1 \\ B_1 \end{pmatrix}, \begin{pmatrix} A_0 \\ B_0 \end{pmatrix}, \begin{pmatrix} A_1 \\ B_1 \end{pmatrix}, \dots$$

- The equation for 2-cycle will be

$$x_{n+2} = x_n \quad y_{n+2} = y_n \quad (5.1)$$

Lecture 8

6.1 1D Flow

- Consider the equation¹

$$\frac{dx}{dt} = f(x) \quad (6.1)$$

Then $x(t)$ is called *flow*

- If $f(x)$ is integrable then

$$\int \frac{dx}{f(x)} = \int dt \quad (6.2)$$

- if $f(x) = \sin x$

$$\begin{aligned} t + c &= \int \operatorname{cosec}(x) dx \\ &= -\log(\operatorname{cosec}(x) + \cot(x)) \end{aligned}$$

- instead of solving the equation; we will plot $\dot{x} = f(x)$ and estimate the properties of the flow
- If $\begin{cases} \dot{x} > 0 : & \text{flow to the right} \\ \dot{x} < 0 : & \text{flow to the left} \end{cases}$
 - denotes the stable fixed point
 - denotes unstable fixed point
 - ◐ denotes half stable
- fixed point is given by $\dot{x} = 0$

¹This occurs in many areas like fluid dynamics, population dynamics, electronics, etc

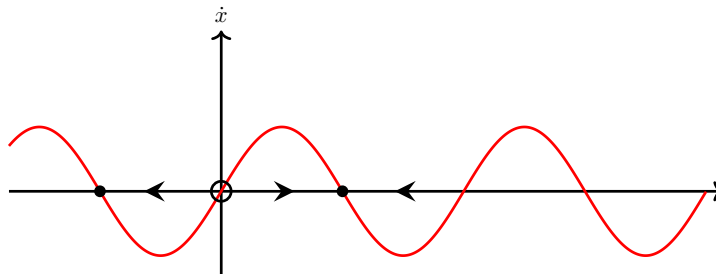


Figure 6.1:

- let $x(t) = x^* + \eta$ for some time t

$$\begin{aligned}
 \frac{dx}{dt} &= f(x^* + \eta) \\
 &= f(x^*) + \eta f'(x^*) + \dots \\
 \frac{d\eta}{dt} &= \eta f'(x^*) \\
 \eta(t) &= \eta(t=0)e^{f'(x^*)t}
 \end{aligned} \tag{6.3}$$

- we define

$$\mu = f'(x^*) \tag{6.4}$$

$\mu < 0 \implies$ stable Fixed Points
 $\mu > 0 \implies$ unstable Fixed Points
 $\mu = 0$ needs extra analysis

Example 6.1. $f(x) = x^{1/3}$ with $x(t) = 0$

$$\begin{aligned}
 \frac{dx}{dt} &= x^{1/3} \\
 \int x^{-1/3} dx &= t + c \\
 \frac{3}{2} x^{-1/3} &= t + c
 \end{aligned}$$

$$c = 0$$

$$x = \left(\frac{2}{3}t\right)^{3/2}$$

6.2 Bifurcation

- We'll now study flows as function of a parameter r

$$\dot{x} = f(x, r) \tag{6.5}$$

Example 6.2. $\dot{x} = r + x^2$

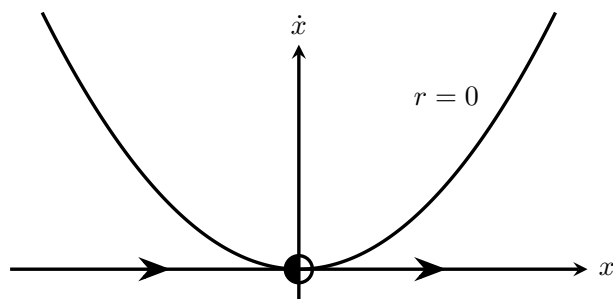


Figure 6.2: half Stable Fixed point at $x = 0$

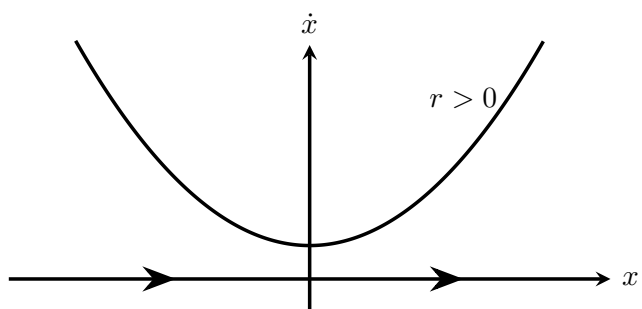


Figure 6.3: No Fixed Point

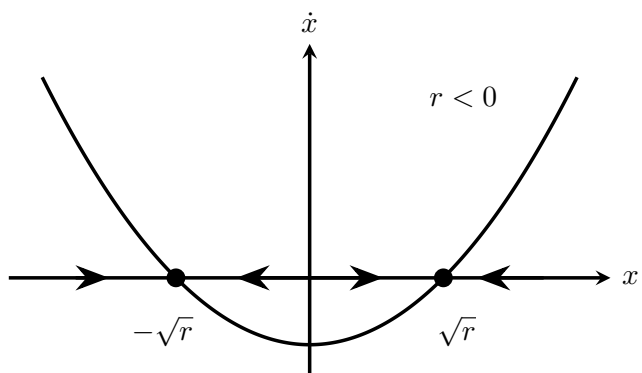


Figure 6.4:

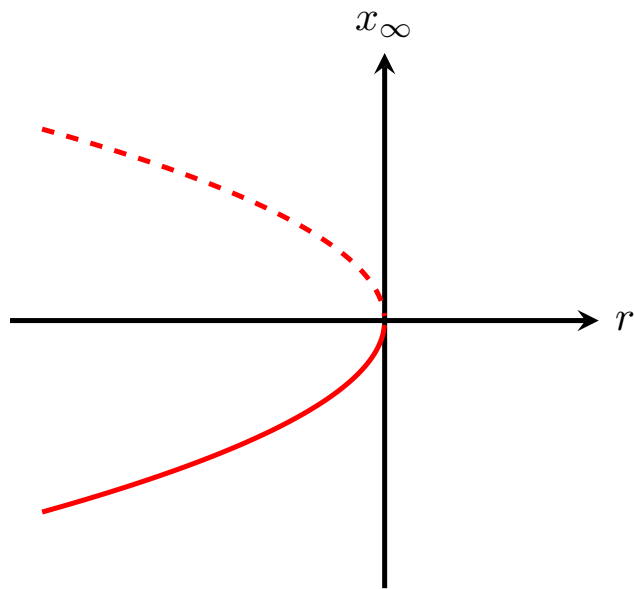


Figure 6.5: Saddle Node Bifurcation

Example 6.3. $\dot{x} = rx - x^2$

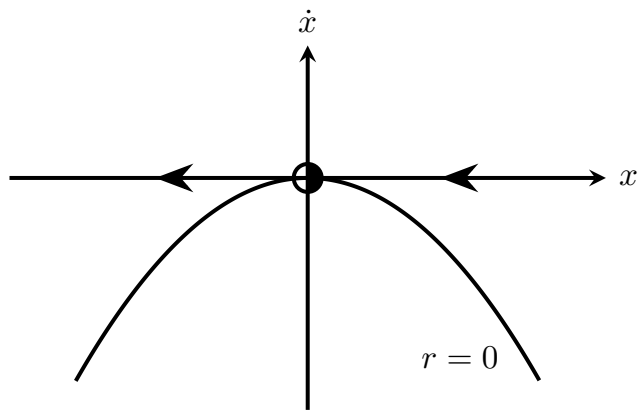


Figure 6.6: